Evolutionary history from genomes to single cells by lossless compression

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Short title: Evolutionary history by lossless compression

One-Sentence Summary: Lossless data compression unifies evolutionary trees, rapid radiations, gene transfer, and convergence across scales of life.

Reconstructing evolutionary history is essential across biology, yet standard pipelines rely on separate models and arbitrary cutoffs to build trees, collapse weak branches, and detect recombination or convergence. Because inheritance copies an ancestor with sparse edits, we recast tree and network inference as

lossless data compression. In Bushwhacker, a branch or lateral link is added only when the shared changes it explains save more information than specifying it costs. Rapid bursts without shared mutations remain unresolved as multiway hubs, horizontal transfers distinguish cleanly from parallel sites, and large phylogenies run orders of magnitude faster than maximum likelihood in far less memory. Across cells, viruses, microbes, and mammals, compression resolves apparent gene conflicts, maps recombinants, disentangles gene conversion from adaptive convergence, and predicts clonal selection.

Evolutionary trees are among the most widely used structures in science: they organize Earth's biodiversity, track unfolding epidemics, trace immune and tumor clones inside individual patients, and anchor comparative genomics across the Tree of Life. Yet even as sequencing has expanded from single genes to millions of genomes, the conceptual foundations used to reconstruct trees have changed remarkably little for nearly half a century. Since distance clustering, parsimony, and probabilistic Markovian likelihood were introduced between the 1960s and early 1980s (*1–3*), major advances have largely focused on scaling tree search and enriching parametric substitution models within those classical paradigms (*4–6*). Genuinely new mathematical perspectives on how evolutionary history can be inferred have been rare.

Here we introduce a different foundation, rooted in information theory: lossless data compression (*7, 8*). Biological inheritance is, at heart, a copying process with sparse edits. Whether an ancestor transmits a genome, a protein, or a somatic cellular profile, related lineages share most of their states because they inherited them from a common source. Recording shared changes once along an ancestral branch is therefore far more economical than recording them separately in every descendant—making an evolutionary graph the natural coordinate system for compressing biological data (Fig. 1). This equivalence is not merely metaphorical: encoding viral and microbial cohorts along an inferred phylogeny physically compresses genomes substantially further than general-purpose algorithms, until pervasive recombination breaks tree-like descent itself.

That insight reframes evolutionary reconstruction around a single question: what is the shortest complete recipe that regenerates every observed genome, protein, or cell from a common ancestor without losing a single character? Starting from one ancestral sequence, our approach builds an evolutionary graph in which each branch transmits only a compact edit list—where changes occurred

and what new states appeared—while every branch or lateral link in the graph adds to the description length. Following Rissanen's Minimum Description Length principle (*7, 8*), every connection must earn its place on one common ledger: a branch or link is added only when the shared edits it eliminates save more information than describing that connection costs. Because description length maps directly onto probability, the information saved by each connection immediately quantifies the odds that it is real—replacing separate model-selection sweeps, pruning cutoffs, and bootstrap resampling with a single self-calibrating score (*9, 10*).

Once every evolutionary claim has to pay its own cost in information, phenomena that normally require incompatible tools fall into place naturally. When cells proliferate, outbreaks super-spread, or species radiate faster than mutations arise, sub-branches have no shared edits to compress; specifying them wastes information, so rapid bursts stay open as multiway hubs (polytomies) rather than inventing unstable binary splits that clash across models and genes (*11–13*). Across uneven rates or billion-year divergences, pinpointing a rare shared change on a quiet ancestral stem saves far more address information than matching chance collisions on fast-evolving branches, defeating long-branch artifacts (*14*) and scaling smoothly to continuous substitution-rate models (*15, 16*). And once vertical descent has compressed everything a tree can explain, residual similarities between distant lineages separate by their genomic footprint: a transferred segment pays just two endpoint coordinates to explain dozens of linked differences—exposing recombination, horizontal gene transfer, and paralog gene conversion (*17, 18*)—whereas adaptive convergence stands out as scattered, high-leverage parallel sites (*19*).

We implemented this framework in Bushwhacker, an open-source engine that reconstructs probabilistically supported trees and lateral networks in a single pass on laptop hardware, generalizes beyond DNA and protein sequences to single-cell somatic mutations and copy-number states, and compiles genome cohorts into compact lossless .bwz archives. Applied to problems where molecular convergence and recombination previously confounded clades (*20–22*)—and spanning eukaryotic radiations, mammalian proteomes (*23*), pandemic viruses, prokaryotic domains (*24, 25*), and single-cell immune and tumor cohorts (*26, 27*)—lossless compression resolves apparent gene-tree conflict, separates gene conversion from adaptive convergence, maps mosaic recombinants, and predicts in vivo clonal selection directly from observed genotypes.

Trees and networks by lossless compression

To measure how concisely an evolutionary graph explains observed sequences or cellular character profiles, Bushwhacker encodes two parts without loss in natural logarithmic units of information (nats; 1 nat $\approx$ 1.44 bits): the graph itself and the state changes along its edges (Fig. 1A, fig. S1, and table S5) (*28*). Starting from one ancestral sequence at the root, each branch records how many positions changed, where they changed, and what new nucleotide, amino-acid, or somatic states appeared, weighted by empirical exchange probabilities (*16*). Resolving a burst into a new ancestral branch avoids repeating shared changes in each descendant, but adds the overhead of specifying which lineages join the clade and encoding the new branch (Fig. 1B). Because shorter code lengths correspond directly to higher probability ($P \propto e^{-L}$), comparing the description length with and without a branch or lateral link (ΔL) gives its local probability support $P_{\text{local}} = 1/(1+e^{-\Delta L})$ without bootstrap resampling: on a four-taxon burst, zero or one shared change loses information ($\Delta L < 0$) and stays open, whereas two to four shared changes yield positive support ($P_{\text{local}} = 77\%$ to 98%).

Measuring site positions and graph connections in the same units also handles uneven evolutionary rates and non-tree events within one code (Fig. 1C,D and Supplementary Text). Specifying where an edit occurred costs far more on a nearly invariant ("quiet") branch than on a mutation-dense one, so moving a shared change onto an ancestral stem from quiet branches saves substantially more information (5.6-fold in Fig. 1C) than grouping chance matches on fast-evolving branches (*2*)—extending smoothly at billion-year divergences to standard empirical exchangeability and fixed eight-category rate variation (*8, 15, 16*) via discrete site weights rather than multi-gigabyte likelihood tables (*28*). Once the vertical tree is refined, candidate lateral links are judged by their genomic footprint: a contiguous transferred block needs only two endpoint coordinates to explain its linked changes, whereas scattered parallel changes must name each mutated position one by one, separating recombination and gene conversion from site-specific convergence.

When trees hold and when bifurcations fail

Before applying this framework to unresolved biological problems, we asked two foundational questions: can lossless compression match established probabilistic methods when evolution genuinely follows a deep tree, and how does it behave when real histories depart from a single strictly

bifurcating tree? Classical deep phylogenies are a stringent test for any combinatorial method because naive change-counting groups fast-evolving lineages by chance collisions on billion-year divergences—the flaw that originally drove phylogenetics toward computationally intensive likelihood models (*2, 14*). Indeed, an unweighted-parsimony version of our engine collapses on deep prokaryotic divisions (Fig. 2A). By contrast, once edits are weighted by evolutionary rate across long-branch benchmarks (*16*) and genome-scale alignments of mammals (*23*), bacteria, and archaea (*25*), lossless compression recovers higher-rank clades alongside fast maximum-likelihood tools (IQ-TREE 2 (*6*) and FastTreeMP (*4*))—matching them on mammalian orders (100%) and bacterial phyla (94.3%) and trailing by a few percentage points on the deepest archaeal splits (tables S2 and S11). Where overall tree distances differ slightly (0.5 to 2.5 percentage points; table S11), two effects operate: on deep bacterial and archaeal concatenations fixed-rate discrete scoring stops short of continuous branch-length likelihood optimization, whereas on mammalian proteins Bushwhacker leaves mutation-poor stems open, trading a small slice of fine-tip recall to halve false splits (4.2% versus 8.4% for IQ-TREE 2). Because rate-weighted scores are evaluated locally without iterative parameter optimization, thousand-species supermatrices finish in about a minute within laptop memory—one to two orders of magnitude faster than maximum likelihood in a small fraction of the memory—making whole-proteome tree and network censuses practical.

Evaluating confidence in individual branches ordinarily multiplies that computational cost by requiring hundreds of trees to be re-estimated on resampled alignments. Across more than $44,000$ branches with known ground truth, the information saved by each branch directly gauges its reliability in a single pass (Fig. 2B and table S3). Branches with decisive compression savings ($P_{\text{local}} \geq 99\%$, nearly 90% of retained splits) are genuine 98.3% of the time. On the remaining borderline splits (P_{local} between 50% and 99%), local posteriors are more optimistic than observed accuracy (41% to 80% true) because P_{local} tests each branch against collapsing that single edge on one optimized tree rather than averaging over competing topologies; high P_{local} therefore pinpoints dependable clades directly, whereas modest values flag branches near the resolution limit.

Even well-calibrated tree builders break down, however, when evolutionary history is not a single binary tree—whether through rapid bursts, genetic exchange, parallel adaptation, or noisy single-cell measurements. Across $37,020$ ground-truth simulations spanning nucleotides, proteins, coalescent radiations, lateral networks, and single cells, evaluating every feature by description length handles

all three regimes natively (Fig. 2C,D, figs. S2 and S3, and tables S2 to S5). When lineages radiate faster than mutations accumulate, strictly bifurcating likelihood fits chance similarities and invents dozens of false splits per burst—even when statistical polytomy tests are enabled (*13*)—whereas compression leaves mutation-free bursts open as multiway hubs (> 94% split precision) and improves downstream coalescent species-tree inference (*29*). When distant clades exchange DNA tracts or evolve parallel adaptive changes, forcing a single tree pulls convergent lineages together (0% sister-clade retention under five shared convergent changes in maximum likelihood), whereas encoding lateral links on the same ledger keeps the vertical tree intact (99.2% clade retention) and separates transferred tracts from parallel sites (90% to 100% recovery on moderate-to-large events; tables S2, S3, and S5) with a 0.75% (3 of 400) false-link rate on null tree-like controls. Finally, on single-cell somatic mutation and copy-number profiles degraded by allelic dropout, native multi-state and ordered-step coding improves clonal precision and copy-number recovery over binary likelihood and binarized coding while calling zero false clones on healthy diploid cells.

Mutation-free branches explain gene-tree conflict

Roughly two-thirds of published single-gene trees clash with genome-wide phylogenies, fueling debates over hybridization and lineage sorting (*11, 12*). However, a major share of these clashes arises without biological discordance: merely changing the substitution model or random starting seed routinely shifts the tree inferred from the exact same alignment (*13*). Both problems stem from enforcing a strict bifurcation rule onto brief, unmutated lineages. When speciation outpaces mutation in an individual locus, short ancestral stems record zero informative changes; leaving those zero-change stems unresolved ($\Delta L \leq 0$, $P_{\text{local}} \leq 50\%$) states probabilistically that the gene cannot distinguish competing branching orders from a coin flip. On a typical mammalian gene (*CHST13*, 1,176 bp; Fig. 3A), enforced bifurcations create eight conflicting splits that bootstrap support does not cleanly resolve: collapsing near-zero branches removes only one conflict, four errors retain 50–75% ultrafast bootstrap support (UFBoot; pulling bushbabies onto rodents at 63% and sloth onto pika at 62%), and a 95% cutoff discards true Catarrhini (93% UFBoot). By leaving the mutation-free placental hub open, Bushwhacker recovers nine true clades ($P_{\text{local}} = 0.82$–1.0) with zero conflicts in 0.02 s (versus 3.8 s for UFBoot; fig. S7).

Across 45,205 genes in 19 eukaryotic radiations, withholding zero-evidence splits removes estimation noise rather than biological signal (Fig. 3B,C and tables S1 and S6). With no reference tree involved, 95% to 96% of branches that flip across substitution models or independent RAxML/IQ-TREE searches on identical alignments carry $\Delta L \leq 0$ and drop out automatically. Measured against published genome-wide phylogenies, withholding $\Delta L \leq 0$ splits eliminates 61.5% of apparent conflicts (74.9% in early birds (*30, 31*) and 82.5% in early animals (*11*)) while retaining 77.2% of shared clades—nearly invariant to how the reference phylogeny itself was built (*32*)—and compressing amino-acid sequences rather than DNA bypasses saturated third-codon positions and base-composition biases in deep plant and insect radiations (*33*).

Crucially, because species trees are themselves inferred from genes, forcing mutation-free loci into strictly bifurcating trees creates a circular trap: thousands of uninformative genes cast random coin-flip votes that destabilize contested deep nodes (*12*) (Fig. 3D). Weighting branches by their positive compression support ($\Delta L > 0$) breaks that circularity—resolving lungfish as sister to tetrapods where unweighted binary gene counts stall in a three-way tie (*34*), uniting turtles with archosaurs > 10-fold, placing ctenophores as the earliest animal lineage under homogeneous rate-category protein models (*35, 36*), and favoring Atlantogenata at the placental root across both summed signal and unweighted locus majorities (*37*). Passing multifurcating Bushwhacker gene trees directly to ASTRAL-IV (*29*) likewise lowers coalescent species-tree error without bootstrap cutoffs across rapid-speciation "anomaly-zone" simulations—where the most frequent binary gene tree disagrees with the true species tree (table S2)—and 11,164 empirical genes spanning animals, plants, fungi, birds (*38*), and mammals (table S1).

Gene conversion and convergence across mammals

Measuring incomplete lineage sorting, paralog gene conversion, and adaptive convergence together across whole proteomes has been out of reach for likelihood pipelines, which cannot scale to thousands of mammal-wide alignments and routinely let unaccounted gene conversion or convergent sites distort single-gene trees. On the outer hair-cell motor gene *Prestin* (*SLC26A5*; 116 mammals), where sonar-linked replacements famously pull echolocating dolphins inside bats on standard protein trees (*19*), Bushwhacker keeps whales inside Cetartiodactyla and bats monophyletic (P_{local} =

1.00) while isolating a lateral convergence link (+3.9 nats, P_{local} = 98.0%) between dolphins and horseshoe bats across four shared motor-domain codons (Fig. 4A and fig. S4). Scaling this analysis to 19, 953 mammalian proteins across up to 882 species (*23*) shows how strongly a single organismal tree organizes the proteome (Fig. 4B, fig. S5, and tables S1 and S7): 85.9% of proteins—and 97.9% above 1, 500 amino acids—compress more tightly on the mammalian species tree than on an unconstrained gene tree. Meanwhile, the share of potential species-tree splits showing MDL-supported protein discordance (19.8% of all splits; 56.5% of resolved splits) rises with sequence length toward the 20.8% coding-DNA discordance baseline, separating incomplete lineage sorting and paralogy from noise.

Without using synteny or paralog labels, genomic footprint separates two lateral processes across those proteins (Fig. 4C). Contiguous paralog gene conversion appears in 6.0% of mammalian proteins (median tract 85 residues), acts largely within mammalian orders (88.3%), and enriches 3.0- to 6.6-fold inside tandemly arrayed gene clusters—including globins, detoxification enzymes, olfactory receptors, zinc-finger arrays, and cadherins or keratins. Scattered parallel amino-acid changes, by contrast, occur in 11.9% of proteins and cross mammalian orders 38.0% of the time (3.3-fold more often than gene conversion), flagging candidate sites under convergent selection.

Those cross-order parallel sites show that ecological adaptation is sharply focal rather than proteome-wide (Fig. 4D and fig. S5C). Addressing long-standing debate over whether sonar drove convergence across thousands of proteins (*20*) or reflects background noise (*21, 39*), echolocating bats, toothed whales, and solenodons show no proteome-wide excess (0.50–0.75× random expectation), nor do marine (1.31×) or subterranean blind (1.47×) mammals; even an apparent excess in ant- and termite-eating mammals traces to giant proteins such as titin and multicopy receptor families (1.35× after length adjustment; table S7). Instead, adaptive convergence targets ecological guilds within shared physiological machinery: although long cochlear proteins accumulate links at length-expected counts (9 of 27 genes for sparse species-tree protein links, 10 with *CIB2*; 19 on free trees), half of those sparse links (P = 0.0011, 3.0× proteome background) directly unite echolocating bat or toothed-whale clades across *CDH23*, *PCDH15*, *LOXHD1*, and *TMC1* alongside *Prestin* (table S7), much as C_4 photosynthesis concentrates parallel changes in catalytic grass enzymes (*40*).

Outbreak hubs, recombinants, and gene transfer

Pathogen evolution routinely defies binary trees: super-spreading transmits identical or single-mutation genomes to dozens of hosts before substitutions accumulate, co-infections drive recombination between divergent strains, and immune escape or drug resistance selects parallel mutations across independent lineages (*18, 22*). Whereas forced binary trees invent arbitrary transmission chains inside acute outbreaks and likelihood recombination scans cannot scale to pandemic cohorts (*10*), Bushwhacker resolves outbreak hubs, recombinational transfers, and convergent mutations together (Fig. 5A,C and tables S1 and S7). Across $10,062$ SARS-CoV-2 and $5,498$ dengue virus genomes, compression preserves $1,426$ SARS-CoV-2 super-spreading hubs of up to 335 descendants alongside urban dengue bursts—nearly doubling on amino-acid sequences once silent nucleotide changes are ignored—while mapping SARS-CoV-2 Spike and Omicron mosaic tracts, natural dengue recombinants (*18*), and HIV-1 recombination breakpoints (*28*).

Encoding genome cohorts into lossless .bwz archives via arithmetic coding also tests directly whether shared descent is the main physical regularity of biological sequences (Fig. 5B and table S9) (*28*). Storing $10,062$ SARS-CoV-2 genomes (300.9 MB raw) requires only 636 KB as a self-contained lossless archive—63.2 bytes per viral genome, 11.5× smaller than gzip -9 and 1.25× smaller than xz -6 (2.06× smaller on sequence data alone)—with consistent gains across influenza, dengue, organelle, and bacterial genomes. Equally telling is when that advantage reverses: across controlled recombination titrations ($K = 0 \rightarrow 32$), mosaic bat sarbecoviruses (0.71×) (*41*), and recombinant HIV-1 (0.65×), pervasive mosaicism erases any single tree and lets general-purpose compressors that copy substrings from multiple donors overtake tree-based coding, making archive size a direct gauge of tree-like versus mosaic evolution.

In prokaryotes, measuring vertical descent and horizontal transfer in the same information units turns qualitative debates over a microbial "Tree versus Web of Life" (*17*) and the Complexity Hypothesis (*24*) into a quantitative budget (Fig. 5D and table S8). In hospital outbreaks of carbapenem-resistant *Klebsiella pneumoniae* (*42*), the 214-kb core chromosome resolves every branch across nine clinical isolates (6/6 splits at $P_{\text{local}} = 1.00$, with lateral transfer contributing just 0.5% of the chromosomal budget), whereas the 882-bp *blaKPC* plasmid carried by those same strains collapses 5 of 6 splits into multifurcating *blaKPC-2* and *blaKPC-3* star hubs separated by

one nucleotide as plasmid transfer outpaces mutation. Across 122 marker proteins in 1,672 archaeal species (*25*), horizontal transfers concentrate in operational metabolic and regulatory genes (11.8%, 12 of 102—including cross-phylum highways between symbiotic DPANN archaea and their hosts (*43*)) and are absent among core ribosomal or RNA polymerase subunits (0 of 20; Fisher $P = 0.21$; table S8). On single-gene microbiome 16S rRNA alignments scaling to 32,768 taxa (*44*), withholding unsupported splits cuts taxonomy-contradicting branches by 19% to 24% while running 4.5× faster than FastTreeMP (table S1).

Clonal selection in single cells

Inside germinal centers and tumors, positively selected cells divide so rapidly that a single founder spawns a star-like burst of daughter lineages before new somatic mutations arise (*45*). Whereas specialized lineage tools recover these bursts only when given cell-abundance counts or hypermutation models (*46*), reconstructing 147 mouse germinal centers (10,429 B cells across 29 mice, 9,352 unique genotypes (*26*)) from deduplicated sequences alone shows that compression geometry reads out Darwinian selection directly (Fig. 6A,B and table S1).

Grouping B-cell genotypes solely by how many mutant daughter lineages (k) radiate from each node predicts a 9.5-fold rise in in vivo clonal expansion from terminal leaves (5.1%) to $k \geq 3$ burst hubs (48.2%; paired Wilcoxon $p = 5.9 \times 10^{-6}$). Hub degree tracks clonal burst abundance within matched mutational-depth strata and shows a weak continuous correlation with higher measured antigen binding across germinal centers, whereas paired within-repertoire enrichment disappears in neutral simulated repertoires (tables S2 and S10). Across animals, somatic mutations arising independently in five or more mice enrich 5.6-fold inside antigen-contacting complementarity-determining regions (64.0% versus 24.1%, $p = 7.5 \times 10^{-5}$) and confer higher binding affinity (Fig. 6A,B).

Extending the same framework to somatic mutation and copy-number profiles in human cancer—with all cell counts and clinical labels withheld—reveals the same selective signature in patient tumors (Fig. 6A,C,D, fig. S6, and table S10). In a single leukemia patient (AML-38-001, 7,016 cells; Fig. 6C), clonal reconstruction in 4.1 ms recovers an early *NPM1* founder splitting into parallel *IDH1* ($k = 3$) and *IDH2* ($k = 4$) burst hubs ($P_{\text{local}} > 0.99$) that independently spawn

FLT3, *KRAS*, *NRAS*, and *PTPN11* resistance sublineages, whereas Markov chain Monte Carlo sampling (471 s) nests *IDH2* sequentially beneath *IDH1*. Across 168 myeloid single-cell cohorts (129 patients, 1.00 million cells (*47*)), inferred burst hubs ($k \geq 3$) carry 6.18-fold more leukemic cells in vivo than terminal leaves ($p = 1.7 \times 10^{-18}$) and capture 90.3% of stepwise copy-neutral loss-of-heterozygosity transitions; across an independent 123-patient leukemia cohort (*27*), branching hubs likewise produce larger descendant clades ($p = 3.0 \times 10^{-15}$) while signaling resistance mutations enrich late (OR = 1.79, $p = 1.9 \times 10^{-3}$) and recur in parallel in 44 patients (Fig. 6A,D).

Finally, across 4, 950 single-cell whole-genome copy-number profiles from ovarian and breast tumors (*48*), ordered step coding compresses 500-kb copy-number genomes 21.9- to 33.5-fold in 22 s and contracts 39%–53% of noisy binary splits while matching binary baseline separation between primary ovarian tumor and ascites subclones at 95.7% purity (fig. S6 and table S10).

Information as a common evolutionary currency

Viewing evolutionary inference as lossless compression shows that tree topology, branch support, evolutionary rates, gene transfer, and adaptive convergence follow from a single principle: shared descent shortens the description of biological variation. Because branches and lateral links are retained only when they compress the data, unresolved hubs become a quantitative biological signal, marking rapid species radiations, epidemic super-spreading, and Darwinian clonal bursts in immunity and cancer without arbitrary cutoffs. Evaluating vertical clades, transferred tracts, and parallel sites on equal terms separates local paralog gene conversion from cross-order convergence and turns debates over microbial gene transfer into genome-wide budgets, while scaling Tree-of-Life phylogenies to standard laptops and web browsers. Because this principle requires only heritable states copied with sparse changes, it extends naturally beyond sequences and somatic profiles to epigenetic marks, CRISPR lineage barcodes, and morphological or linguistic characters wherever history leaves a compressible trace.

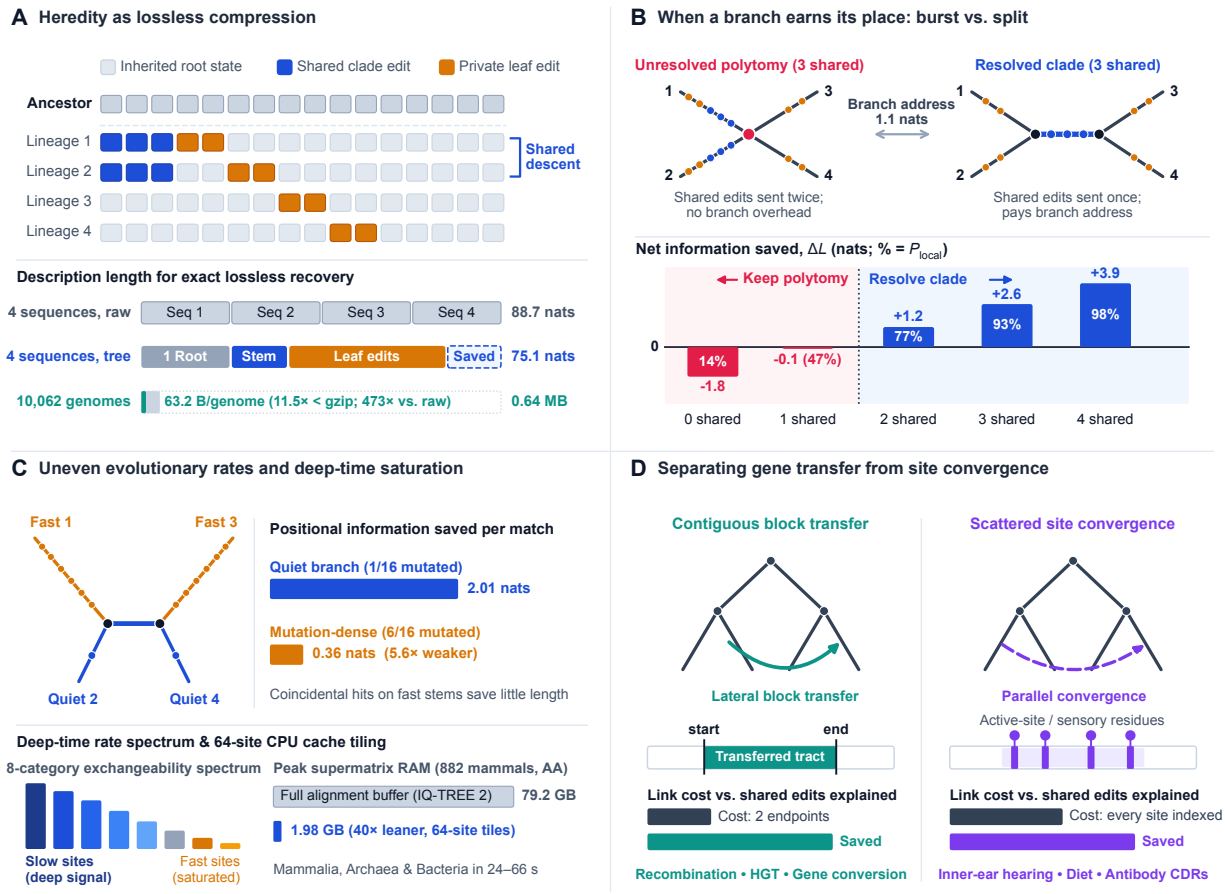


Figure 1: **Evolutionary trees and networks are lossless compression structures.** (**A**) Encoding one ancestral root plus sparse branch edits shortens four 16-site sequences from 88.7 to 75.1 nats (1 nat $\approx$ 1.44 bits), compounding to 63.2 bytes per 29.9-kb genome (11.5× smaller than gzip -9; table S9) across 10,062 SARS-CoV-2 genomes. (**B**) Resolving a burst shares edits across descendants (top) but pays the topological split address (1.1 nats on a quartet). Zero or one shared change loses information (-1.8 and -0.1 nats; $P_{\text{local}} = 14\%$ and 47%) and remains a polytomy, whereas two to four shared changes yield positive net savings (+1.2 to +3.9 nats; $P_{\text{local}} = 77\%$ to 98%). (**C**) Positional coding yields 5.6× greater savings on quiet stems than on mutation-dense branches, while eight-category rate coding in 64-site CPU tiles scales to billion-year supermatrices in under 2.8 GB. (**D**) Contiguous lateral transfers require only two genomic endpoints to explain linked edits, filtering out scattered homoplasy.

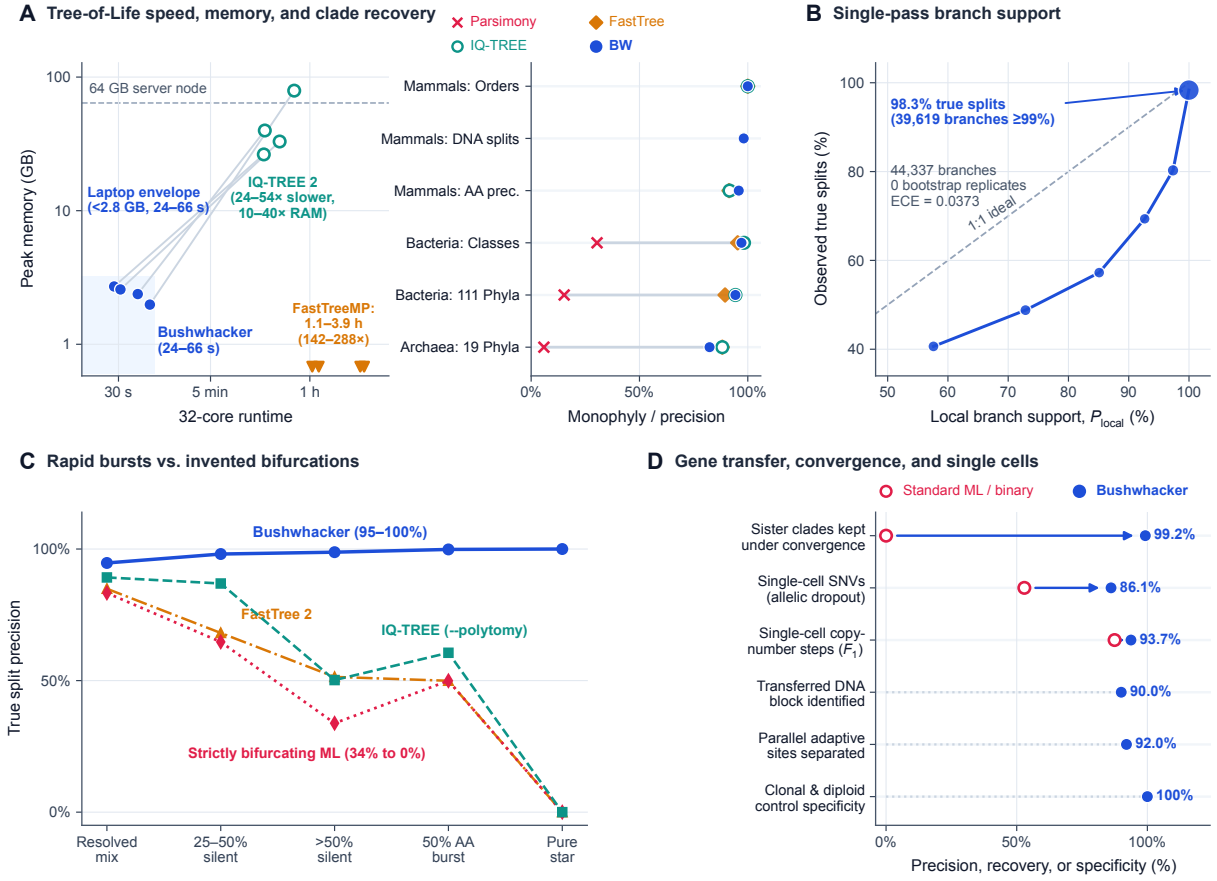


Figure 2: **When trees hold and when bifurcations fail across genomes, networks, and single cells.** (**A**) Thirty-two-core runtime and peak memory across four Tree-of-Life supermatrices (882 mammals, DNA and protein; 1,672 archaea; 1,000 bacteria; Bushwhacker, 24–66 s and 1.98–2.71 GB versus 26–79 GB for IQ-TREE 2; triangles, FastTreeMP) alongside clade monophyly and split precision against IQ-TREE 2, FastTreeMP, and unweighted parsimony (tree distance within 0.5–2.5 percentage points; table S11). (**B**) Bootstrap-free local branch support (P_{local}) versus observed split accuracy across 44,337 branches (98.3% true at $P_{\text{local}} \geq 99\%$, 40.6%–80.3% across intermediate bins; circle area proportional to branch count; table S3). (**C**) Split precision across 36,120 simulated nucleotide, protein, and coalescent histories (0.00–0.18 versus 23–61 false splits per rapid burst; table S2). (**D**) Clade retention, event recovery, and control specificity across 800 benchmarks of horizontal transfer, parallel convergence, and single-cell somatic states (400 null controls, 3/400 false shunts; tables S2–S5).

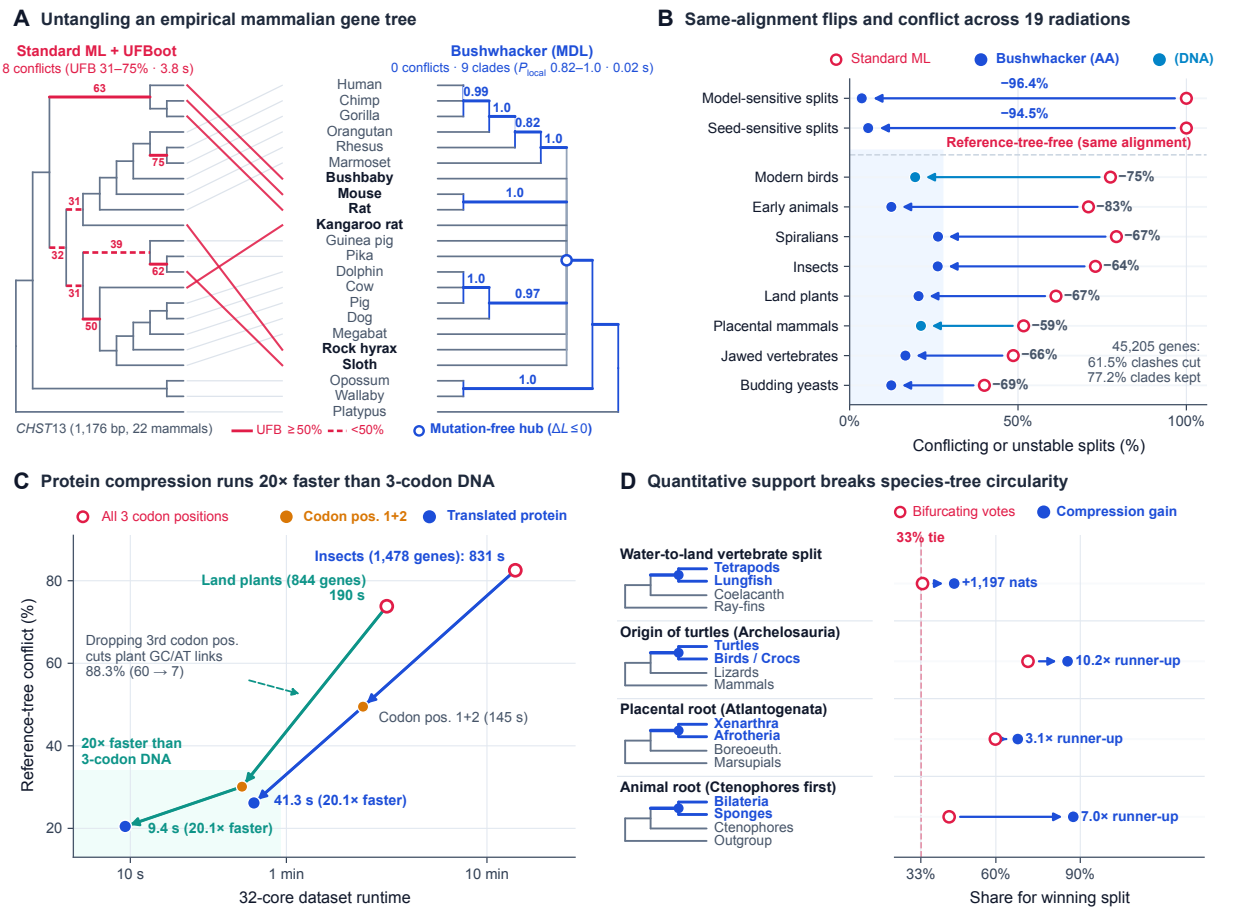


Figure 3: **Mutation-free stems account for widespread gene-tree conflict and model sensitivity.** (**A**) Mammalian *CHST13* tanglegram (1, 176 bp, 22 species) contrasting standard maximum likelihood annotated with ultrafast bootstrap support (8 conflicting splits in crimson: solid ≥ 50%, dashed < 50% UFBoot; 3.8 s on 1 core; fig. S7) against Bushwhacker (`-empirical-alpha`: 0 conflicts; 9 true clades at P_{local} = 0.82–1.0 in 0.02 s; open circle, $\Delta L \leq 0$ placental hub). (**B**) Reference-free controls showing 96.4% removal of model-sensitive splits (7, 200 genes) and 94.5% removal of seed-sensitive splits across 20 RAxML/IQ-TREE runs per locus (83, 180 trees across 4, 159 alignments; top; table S6), above conflict reduction across 45, 205 genes in 19 eukaryotic radiations (61.5% clashes removed; 77.2% shared clades retained; bottom). (**C**) Runtime and conflict across coding DNA, codons 1 + 2, and translated proteins in plants (844 genes) and insects (1, 478 genes), cutting saturated plant links by 88.3% (60 to 7). (**D**) Competing quartet vote shares before (strictly bifurcating counts) and after ($\Delta L > 0$ weighting) breaking species-tree circularity (dashed line, 33% three-way tie).

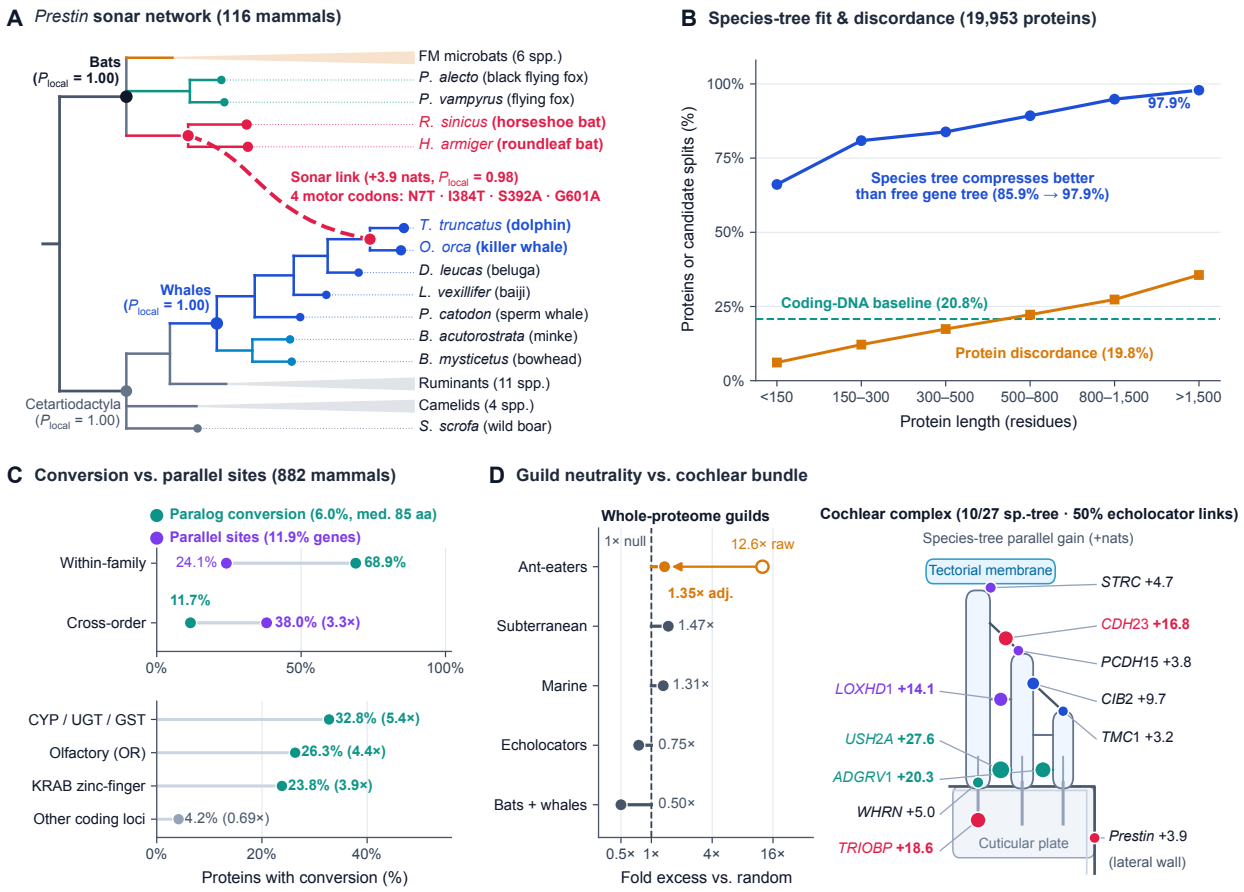


Figure 4: **Unified mammalian census separates organismal inheritance, gene conversion, and convergence.** (**A**) Mammalian *Prestin* (*SLC26A5*, 116 species, 2,235 bp; Laurasiatheria shown) preserves monophyletic bats and Cetartiodactyla ($P_{\text{local}} = 1.00$) while isolating the echolocator convergence link (+3.9 nats, $P_{\text{local}} = 98.0\%$) across four motor-domain codons (N7T, I384T, S392A, G601A). (**B**) Proteins compressing better on the species tree than a free gene tree (85.9% overall; 97.9% above 1,500 residues, blue) and species-tree split discordance ($\Delta L > 0$: 19.8% of all splits overall, amber; 20.8% coding-DNA baseline, teal) across 19,953 TOGA2 proteins in 882 mammals. (**C**) Separation of 8,670 species-tree links into contiguous paralog conversion (6.0% of proteins, 2,251 tracts, median 85 residues; 68.9% intra-family, 3.0–6.6× enriched in tandem clusters) and scattered parallel-site links (11.9%, 6,419 links; 38.0% cross-order). (**D**) Guild-wide neutrality (0.50–1.47×; ant-eating 1.35× length-adjusted; left) versus focal echolocator convergence in cochlear proteins (10/27 species-tree, 19/27 free-tree; 50% of sparse links unite echolocator clades, right; table S7).

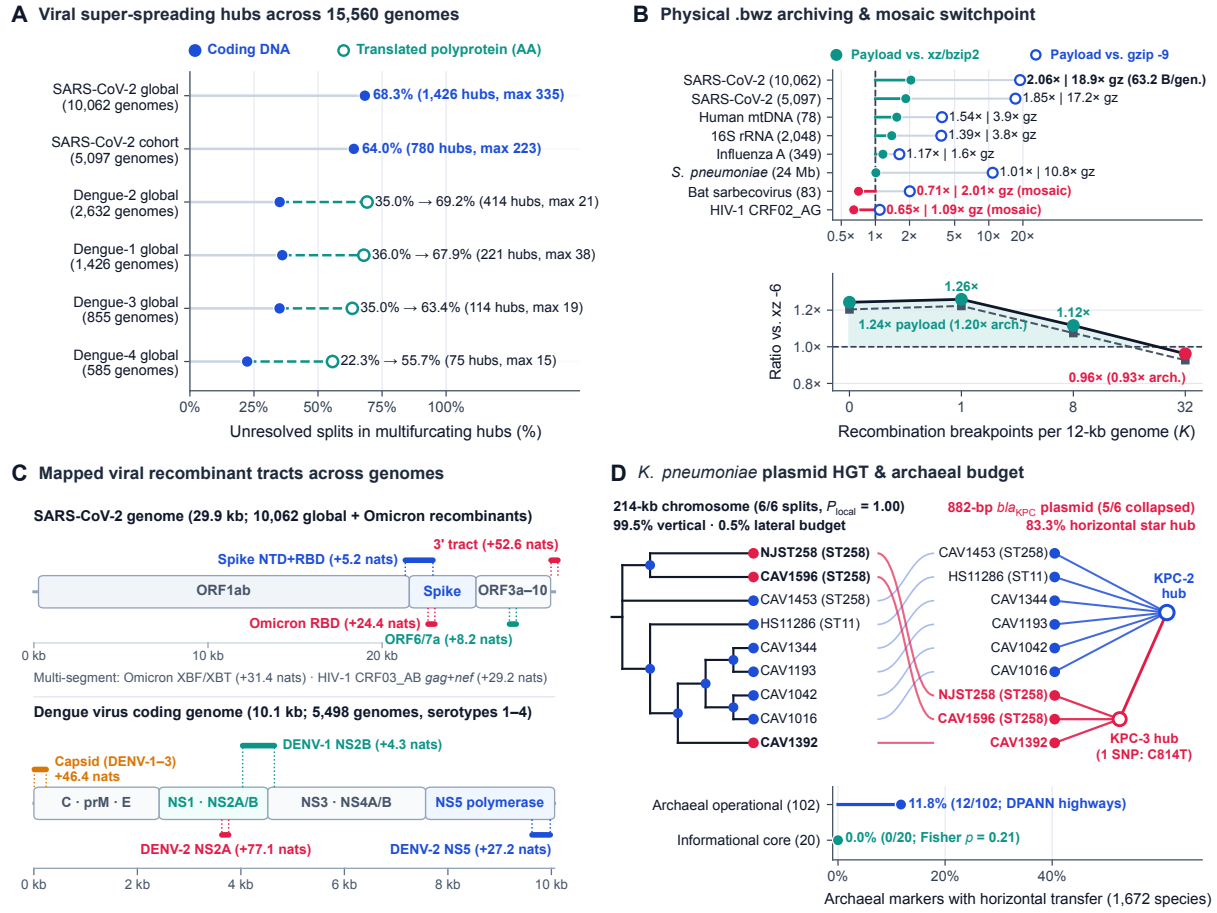


Figure 5: **Viral outbreaks, physical genome archiving, recombination, and prokaryotic gene transfer budgets.** (**A**) Multifurcating outbreak hubs and largest burst sizes across 10, 062 global SARS-CoV-2 (68.3% unresolved; 1, 426 hubs up to 335 descendants) and 5, 498 dengue genomes (serotypes 1–4) in coding DNA versus translated polyproteins. (**B**) Lossless .bwz sequence payload and full archive reductions over xz -6 or bzip2 -9 (up to 2.06× payload) and gzip -9 (up to 18.9×; 38.5 B/genome payload, 63.2 B/genome archive on SARS-CoV-2; top), compared against crossover below 1× under breakpoint titrations ($K = 0 \rightarrow 32$; bottom), bat sarbecoviruses (0.71×), and recombinant HIV-1 (0.65×). (**C**) Mapped mosaic tracts along the 29.9-kb SARS-CoV-2 and 10.1-kb dengue genomes. (**D**) Nine carbapenem-resistant *Klebsiella pneumoniae* isolates across the 214-kb chromosome (6/6 splits, 99.5% vertical budget) versus the 882-bp *blaKPC* plasmid (5/6 splits collapsed into C814T star hubs; top), and archaeal transfer rates across 122 markers in 1, 672 species (11.8% operational versus 0.0% informational, Fisher $p = 0.21$; bottom).

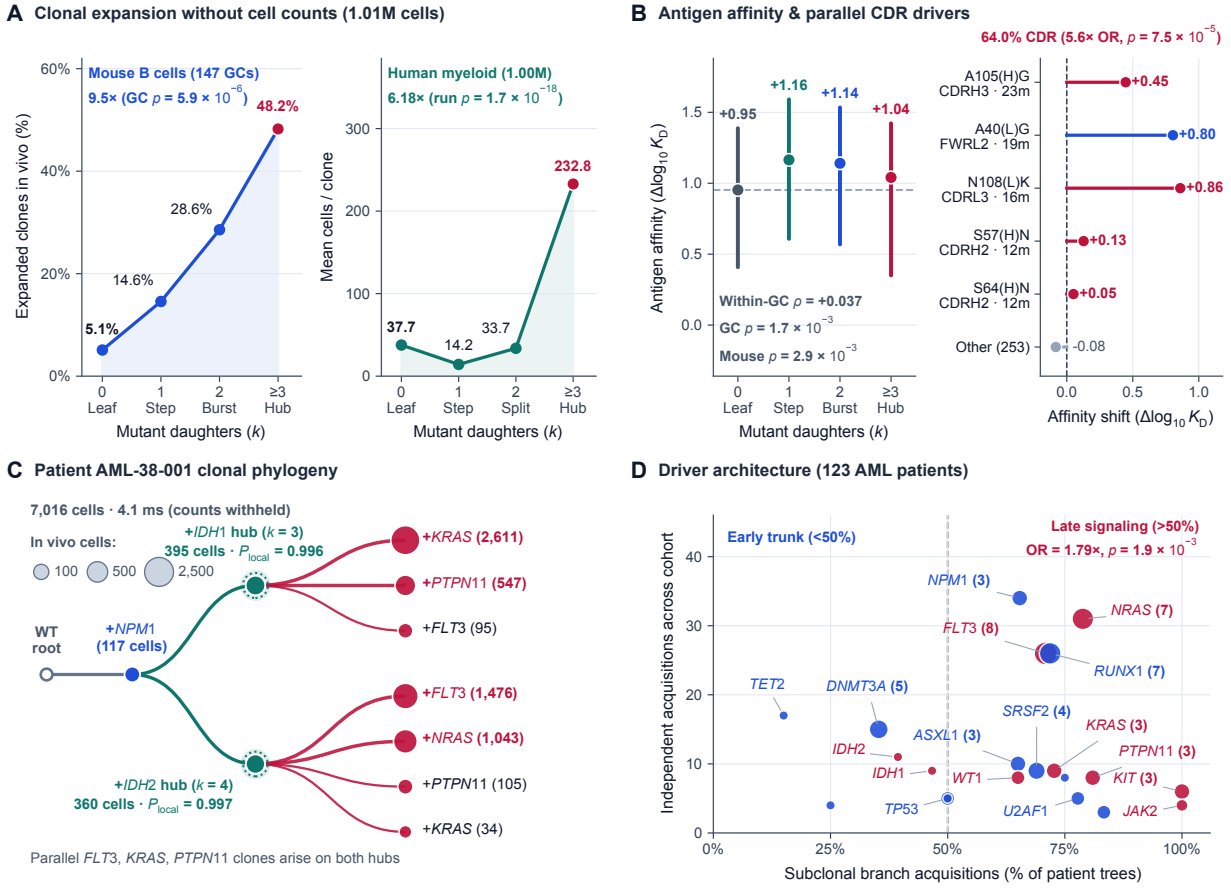


Figure 6: **Somatic compression geometry detects clonal selection across immune and cancer cells.** (**A**) With cell counts withheld, mutant daughter degree k predicts in vivo clonal expansion across 147 mouse germinal centers (10,429 B cells; 5.1% → 48.2%, 9.5×, paired Wilcoxon $p = 5.9 \times 10^{-6}$; left) and 168 myeloid single-cell cohorts (1,001,609 cells; 37.7 → 232.8 cells/clone, 6.18×, $p = 1.7 \times 10^{-18}$; right). (**B**) Antigen affinity ($\Delta \log_{10} K_D$) across B-cell hubs (within-GC $p = 1.7 \times 10^{-3}$, within-mouse $p = 2.9 \times 10^{-3}$; left) and cross-mouse parallel mutations (64.0% in complementarity-determining regions versus 24.1% background, 5.6× odds ratio, $p = 7.5 \times 10^{-5}$; right). (**C**) Patient AML-38-001 phylogeny (7,016 cells; 4.1 ms without counts; circle area proportional to held-out cell count): an *NPM1* trunk splits into *IDH1* ($P_{\text{local}} = 0.996$, $k = 3$) and *IDH2* ($P_{\text{local}} = 0.997$, $k = 4$) resistance hubs. (**D**) Early epigenetic versus late signaling drivers across 123 AML patients (OR = 1.79, Fisher $p = 1.9 \times 10^{-3}$; counts denote within-patient recurrences).

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Acknowledgments

We thank collaborators and open-data consortia for making multi-scale genomic and single-cell cohorts publicly available. Google Gemini large language models (Gemini 4 and Gemini 3.8 Flash, accessed via Google Antigravity) were used under author supervision to assist with software, web-application, and plotting-script engineering, refactoring, and manuscript copyediting; the human authors independently verified and take full responsibility for all mathematical derivations, code, analyses, figures, and citations.

Funding: This work was supported in part by the US National Institutes of Health (grants R01GM151683, R01GM144468, U24AI183870, and R01HG009299 to S.L.K.P. and A.N.) and the US National Science Foundation (grants DBI-2419522 and DEB-2216289 to S.L.K.P.).

Author contributions: Conceptualization: S.L.K.P., D.P.M., A.N. Methodology and software: S.L.K.P., D.P.M., S.W., S.J.K.P. Investigation, validation, and visualization: S.L.K.P., D.P.M., S.W., D.C., S.J.K.P., A.N. Writing—original draft and review & editing: S.L.K.P., D.P.M., S.W., D.C., S.J.K.P., A.N.

Competing interests: There are no competing interests to declare.

Data, code and materials availability: The open-source Rust and WebAssembly implementation of Bushwhacker (*49*), the .bwz lossless archival codec, interactive browser application (`https://veg.github.io/bushwhacker/`), reproducibility scripts, and empirical and ground-truth validation tables spanning more than 81, 000 empirical alignments, 1.01 million single cells, and 37, 020 simulated evolutionary histories are archived at `https://github.com/veg/bushwhacker` and bundled in the Zenodo v1.0 replication archive (`bushwhacker_science_zenodo_v1.0`; see (*28*) for accessions and DOIs). No physical materials were generated in this work.

Supplementary materials

Materials and Methods

Supplementary Text

Figs. S1 to S7

Tables S1 to S11

References *(50–91)*